

Area Moment of Inertia

1. General Integral Form: $I_x = \int_A y^2 dA$ $I_y = \int_A x^2 dA$

2. "Area" vs. "Mass" Moment of Inertia: Similar mathematically, but used for entirely different purposes in mechanics.

(a) "Mass" MOI: $I = \int_m r^2 dm$ ~ mass element = $\rho \cdot dV$

Used in rotational dynamics, $\underbrace{\sum M_G}_{\text{Net Torque}} = I_G \underbrace{\alpha}_{\text{Angular Accel.}}$

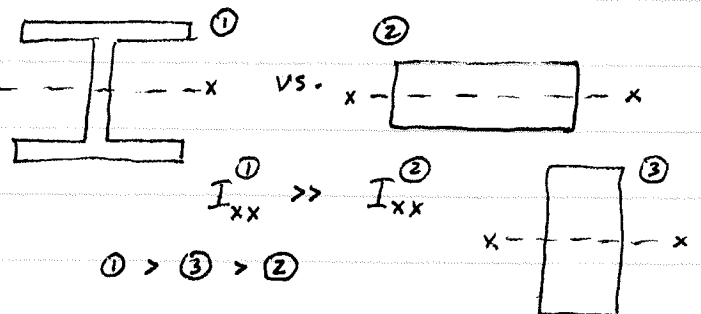
I_G is a body's resistance to angular acceleration.

(b) "Area" MOI: Used in Mechanics of Materials when calculating stresses in beams, beam deflection under loads... Stresses, angle of twist in shafts in torsion.

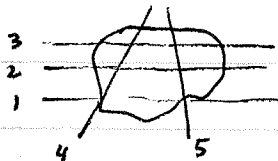
3. Properties of the Area Moment of Inertia...

(a) $I = f(\text{distance}^2)$

At right, ① and ② have same area, but matⁿ distributed differently....



(b) I is different about every axis passing thru an area...

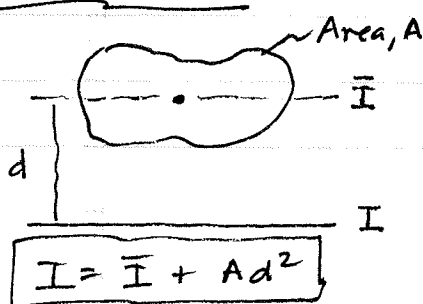


$I_1 \neq I_2 \neq I_3 \neq I_4 \neq I_5$ in general...

(c) If know I about one axis, can use the parallel axis theorem "PAT" to find I about parallel axes....

NOTE! One of these axes must be a centroidal axis.

PARALLEL AXIS THEOREM



(d) Note that $\bar{I}$ is a minimum for an area (at centroid).

As move away from centroidal axis, add "transfer term" = Ad^2

3. Properties of the Area Moment of Inertia (cont'd)

(e) We are most interested in the centroidal MOI's

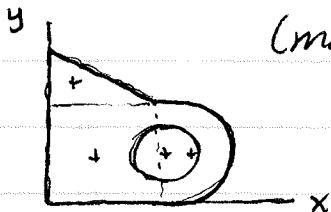
(I usually denote these as $\bar{I} \dots$)

but MOI's along $x + y$ axes are easier to calculate

4. Area Moment of Inertia for Composite Areas

(made up of sub-areas of known shapes)

Use Parallel Axis Theorem (PAT):



$$I_x = \sum (\bar{I}_x + A d_y^2)$$

$$I_y = \sum (\bar{I}_y + A d_x^2)$$

Do not use I_{base} terms here;
must use centroidal values

Area MOI's for Common Shapes

